

Fast Clustering of Retired Lithium-ion Batteries for Secondary Life

with A Two-step Learning Method

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Experimental procedures

We have purchased 507 cylindrical 18650 batteries $\text{LiNi}_{1-x-y}\text{Co}_x\text{Mn}_y\text{O}_2$ (NCM)/graphite from a different commercial company. These batteries were charged and discharged at 1 A under a controlled temperature environment at 25°C. All LIBs were charged using the manufacturer's recommended method (i.e., a constant current at 0.5 A followed by a constant voltage at 4.2 V with C/200 or 1/2h cut off) during the conditioning. Besides the aging cycle, after charging the LIBs to full charge with 1 A, we use 0.5 A to discharge and obtain its discharge capacity, which is regarded as its tested capacity and our benchmark for estimation accuracy. Then we charge the battery with 1 A for 6 minutes and let it rest for 30 minutes to make the voltage more stable. Next, charge with 0.5 A for 10 seconds, rest for 30 seconds, discharge for 10 seconds under the same current of 0.5 A, and rest for 30 seconds. After that, we use the current of 0.6 A, 0.7 A, 0.8 A, 0.9 A, 1.0 A, 1.25 A, 1.5 A, 1.75 A, and 2.0 A to repeat the charge and discharge each time.

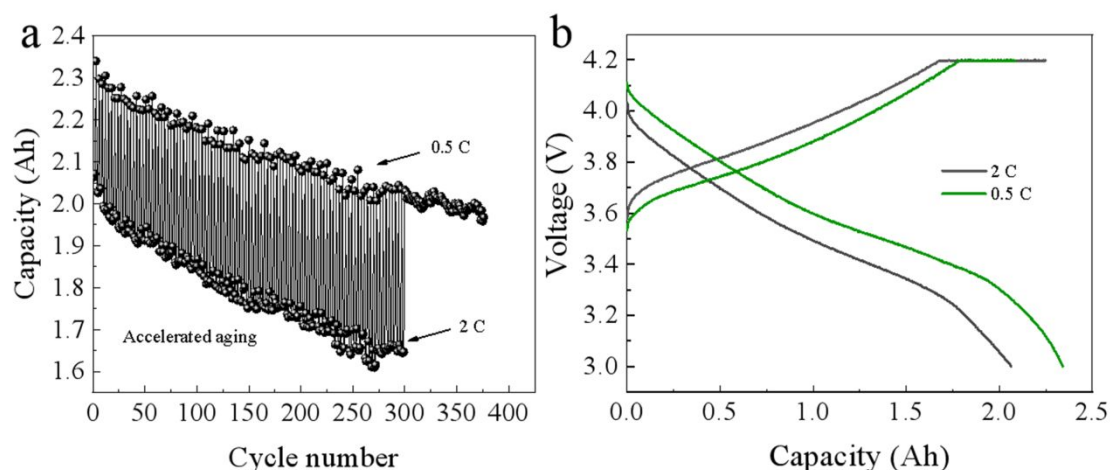


Fig. S1. (a) Capacity fading curve of a typical lithium-ion batteries at accelerated testing. (b) Charging and discharging curve at different current rate.

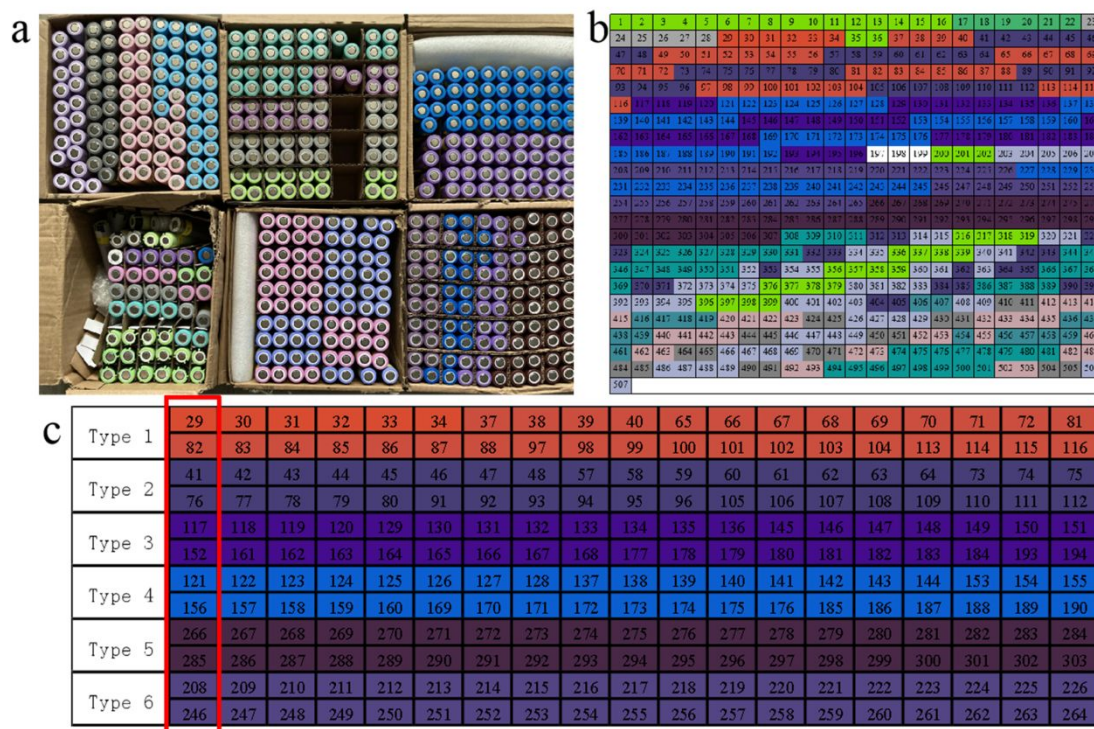


Fig. S2. All the retired lithium-ion batteries tested in this experiment. (a) 507 (18650 Cylindrical NCM Battery). (b) The corresponding types distribution (the same color indicates the same types). (c) Six types of LIBs selected and the battery inside the red frame is used for thermal measurement.

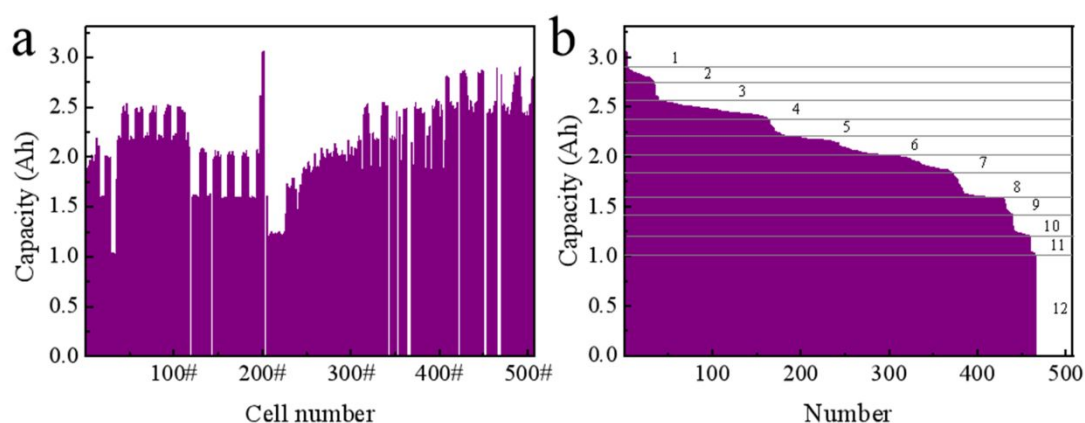


Fig. S3. (a) The discharge capacity distribution of 507 batteries collected by the proposed method and (b) the arrangement of from large to small.

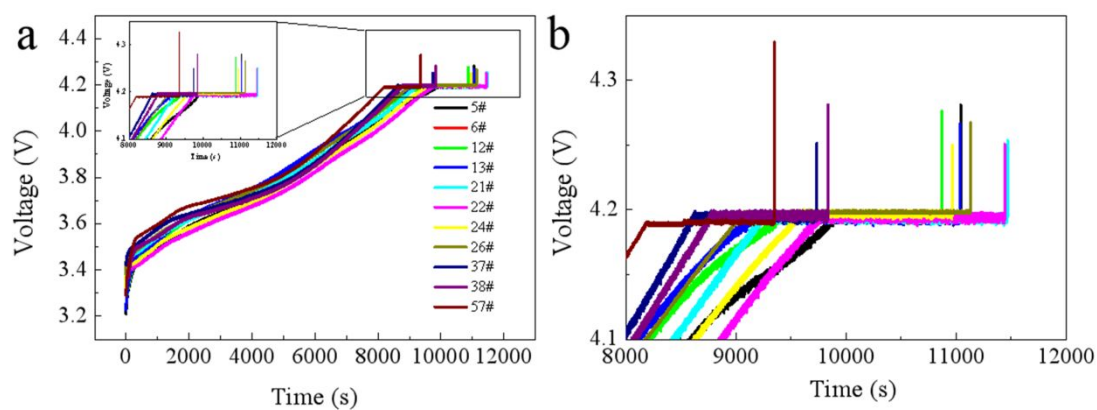


Fig. S4. Abnormal batteries in Cluster 12.

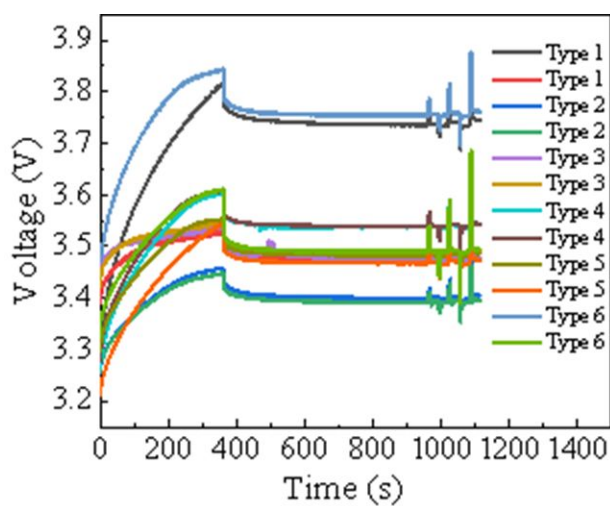


Fig. S5. Charging profile of different types of LIBs of pulse test.

Algorithm The K-means Algorithm

Input: The data set $X = [x_1, x_2, \dots, x_N]^T$ and the number of clusters K ;

Output: The clustering result.

1. Randomly choose K samples to be the initial cluster centers and define the center as $C = [c_1, c_2, \dots, c_K]^T$;
 2. Set the cluster change sign $sign = 1$;
 3. **while** $sign = 1$ **do**
 4. **for** each $x_i \in X$ **do**
 5. Find the smallest distance $d(x_i, c_j) = \|x_i - c_j\|_2$ between x_i and $c_j \in C$;
 Let $j = \arg \min d(x_i, c_j)$, assign x_i to cluster j , denoted as $Cluster_j$;
 6. **end for**
 7. Update the center C for each $c_j \in C$ according to $c_j = \frac{1}{N_j} \sum_{x \in Cluster_j} x$,
 where N_j denotes the cardinality of $Cluster_j$
 8. **if** C does not change **then**
 9. Set $sign = 0$;
 10. **end if**
 11. **end while**
 12. Output the clustering result.
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Fig. S6. The K-means algorithm.

Algorithm The GMM Algorithm

Input: The data set $X = [x_1, x_2, \dots, x_N]^T$ and the number of clusters K ;

Output: The probability that the observation data X belongs to the cluster C , denoted as P ; mean(μ) and variance(Σ) of Gaussian distribution of each cluster.

13. **E-step:** Denote the clusters vector as $C = [c_1, c_2, \dots, c_K]^T$, the latent variables as $Z = [z_1, z_2, \dots, z_N]^T$. The expectation of log-likelihood of complete-data (X, Z) is $Q(\theta, \theta^{(t)}) = \mathbb{E}_{Z|X, \theta^{(t)}} [\ln p(X, Z | \theta)] = \sum_{k=1}^K \sum_{i=1}^N [\ln p_k + \ln N(x_i | \mu_k, \Sigma_k)] \cdot p(z_i = c_k | x_i, \theta^{(t)})$ where θ is the parameters to be estimated, which contains $P = [p_1, p_2, \dots, p_K]^T$, $\mu = [\mu_1, \mu_2, \dots, \mu_K]^T$ and $\Sigma = [\Sigma_1, \Sigma_2, \dots, \Sigma_K]^T$.
14. Set the iteration sign $sign = 0$;
15. **while** $sign = 0$ **do**
16. **M-step:** Maximize the expectation to update θ : $\theta^{(t+1)} = \arg \max_{\theta} Q(\theta, \theta^{(t)})$. Specifically, take the derivative of $Q(\theta, \theta^{(t)})$ with p_k, μ_k and Σ_k , respectively, and then set them equal to zero, we obtain:
- $$p_k^{(t+1)} = \frac{1}{N} \sum_{i=1}^N p(z_i = c_k | x_i, \theta^{(t)})$$
- $$\mu_k^{(t+1)} = \frac{\sum_{i=1}^N p(z_i = c_k | x_i, \theta^{(t)}) x_i}{\sum_{i=1}^N p(z_i = c_k | x_i, \theta^{(t)})}$$
- $$\Sigma_k^{(t+1)} = \frac{\sum_{i=1}^N p(z_i = c_k | x_i, \theta^{(t)}) (x_i - \mu_k^t)(x_i - \mu_k^t)^T}{\sum_{i=1}^N p(z_i = c_k | x_i, \theta^{(t)})}$$
17. **if** t reaches the number of iterations of convergence **then**
Set $sign = 1$;
- else** $t = t + 1$
18. **end if**
19. **end while**
20. Output the clustering result.
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Fig. S7. The GMM algorithm.

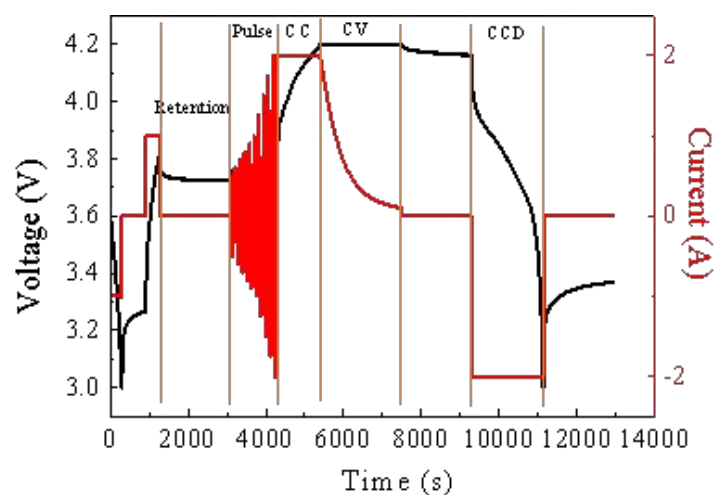


Fig. S8. Current and voltage changes during thermal measurement.

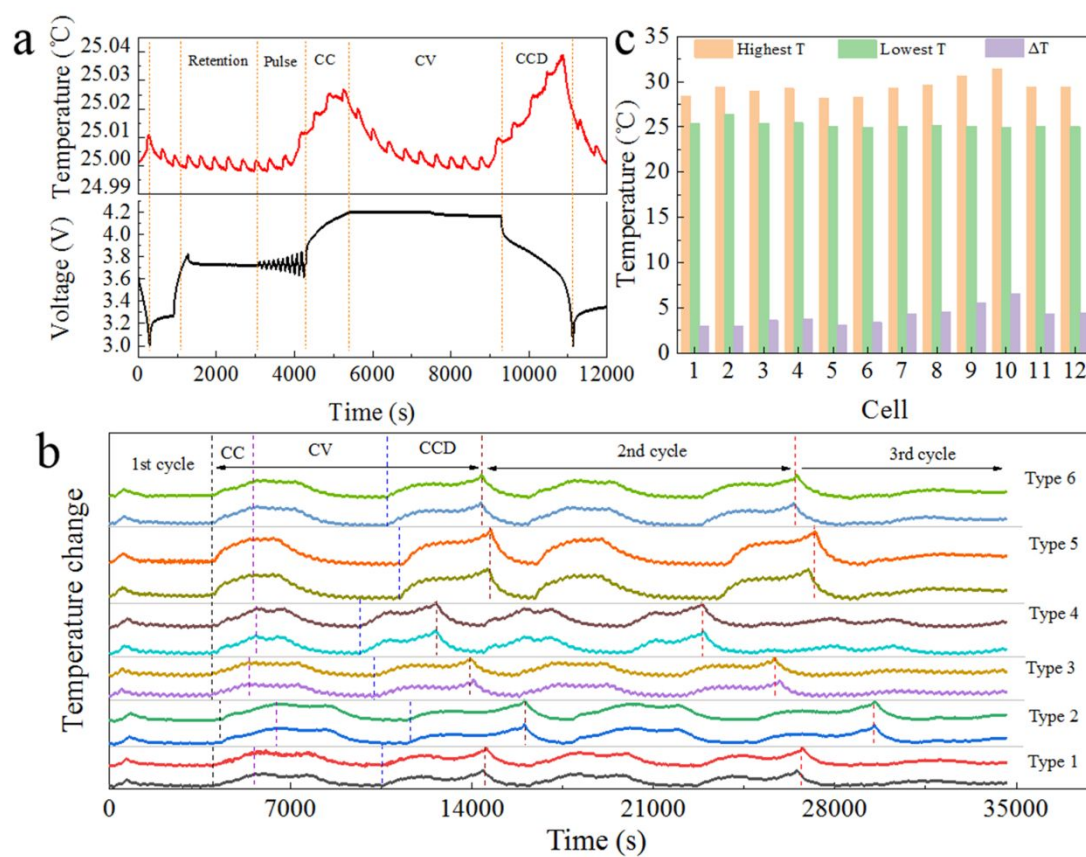


Fig. S9. Thermal measurement results. (a) Typical temperature changes of a single battery with voltage and time. (b) Temperature change of six different types of LIBs. (c) Summary of temperature changes of six types of batteries.

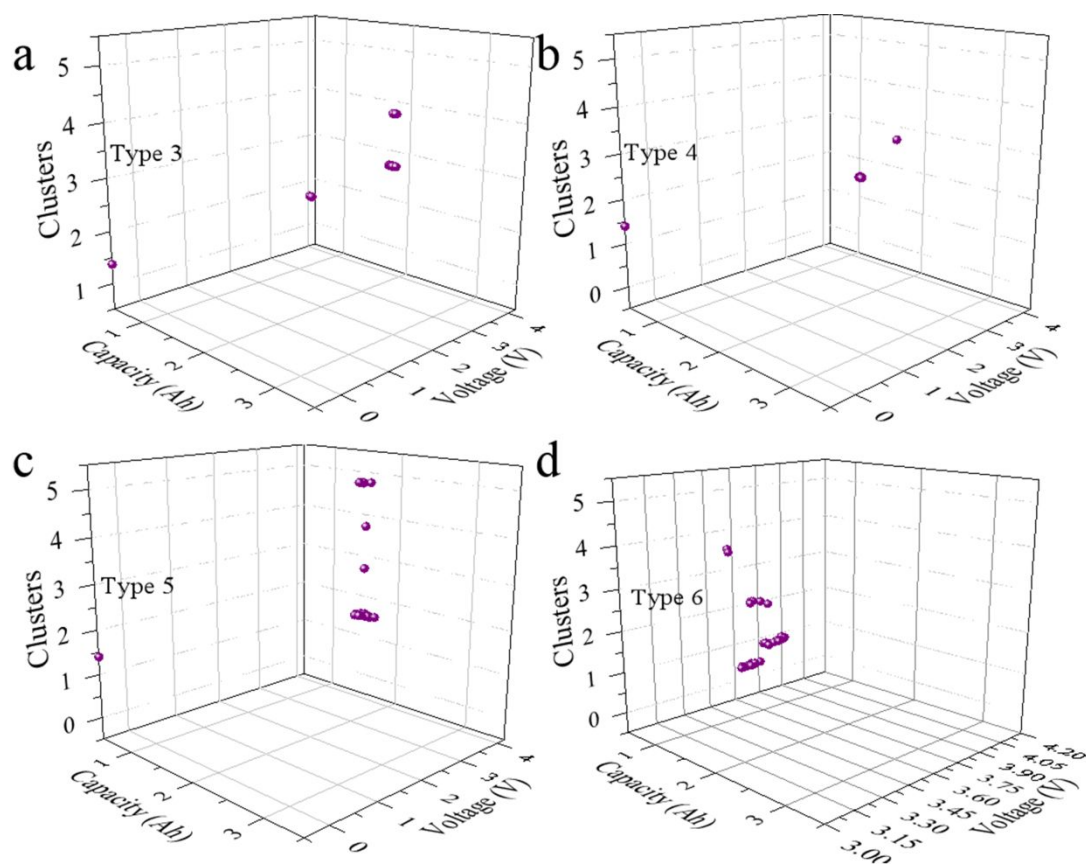
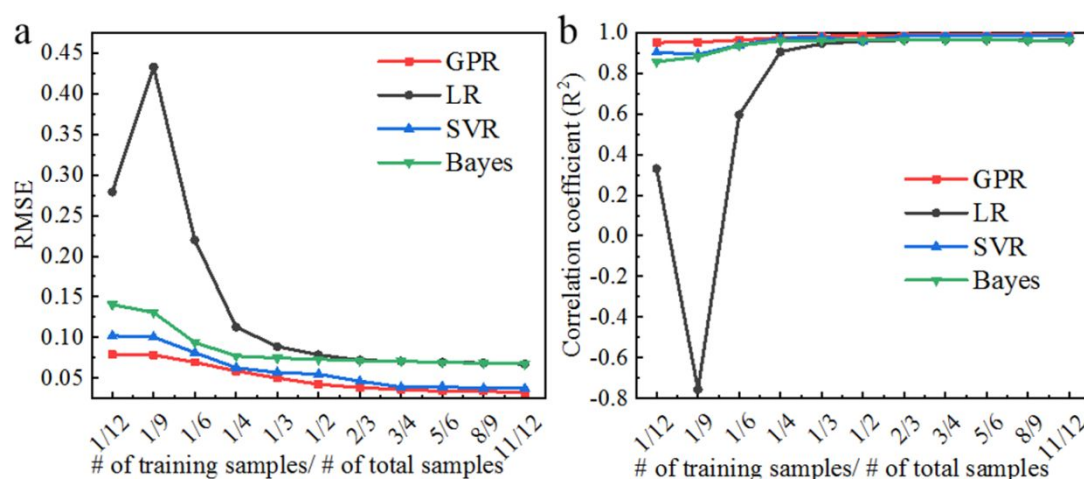


Fig. S10. Clustering results of (a) Type 3, (b) Type 4, (c) Type 5 and (d) Type 6 by K-means.

Table S1. Clustering results of Type 3, 4, 5 and 6.

	Cluster	Cell number	OCV (V)	Cell sorted for capacity (Ah)
Type 3	1	3	0	0
	2	1	3.252	1.577
	3	24	3.387 ~ 3.427	1.948 ~ 2.091
	4	10	3.382 ~ 3.428	2.044 ~ 2.091
Type 4	1	2	0	0
	2	35	3.192 ~ 3.251	1.58 ~ 1.628
	3	1	3.398	2.025
Type 5	1	2	0	0
	2	1	3.425	1.919
	3	28	3.12 ~ 3.302	1.943 ~ 2.172
	4	6	3.112 ~ 3.214	2.0 ~ 2.106
	5	1	3.32	1.987
Type 6	1	19	3.154 ~ 3.393	1.186 ~ 1.262
	2	2	3.235 , 3.275	1.423 , 1.564
	3	11	3.304 ~ 3.428	1.804 ~ 1.953
	4	6	3.229 ~ 3.280	1.84 ~ 2.035

**Fig. S11.** RMSE and correlation coefficient (R^2) of different proportions of training samples to total samples under four types of regression method.

The Linear Regression (LR), Support Vector Regression (SVR), Bayesian Ridge Regression (Bayes) and Gaussian Process Regression (GPR) are compared to explore the effectiveness of each regression method. As shown in **Fig. S11**, considering low RMSE (**Fig. S11(a)**) and high correlation coefficient t (**Fig. S11(b)**), among which GPR significantly outperformed the other methods. The detailed data are shown in **Table S2**, **S3** and **S4**. Compared to LR, SVR, Bayes and GPR were more effective. GPR has a significant advantage when the ratio of training samples to total samples is less than 1/3. Therefore, GPR is the best machine learning model for battery capacity estimation.

Table S2. Details of training and testing samples, performances of capacity estimation models of Linear Regression.

Training samples/total samples	Number of training samples in every type	Number of training samples	Number of testing samples	RMSE	R ²
1/12	3	18	200	0.278963	0.330396
1/9	4	24	194	0.432346	-0.75553
1/6	6	35	183	0.219666	0.595439
1/4	9	56	162	0.112832	0.905106
1/3	12	73	145	0.088771	0.945576
1/2	18	108	110	0.078763	0.956696
2/3	25	145	73	0.072219	0.962986
3/4	28	162	56	0.070785	0.963288
5/6	31	183	35	0.069612	0.96387
8/9	33	194	24	0.068763	0.961204
11/12	34	200	18	0.067425	0.961965

Table S3. Details of training and testing samples, performances of capacity estimation models of Support Vector Regression.

Training samples/total samples	Number of training samples in every type	Number of training samples	Number of testing samples	RMSE	R ²
1/12	3	18	200	0.102214	0.902518
1/9	4	24	194	0.100601	0.891928

1/6	6	35	183	0.081514	0.938061
1/4	9	56	162	0.062783	0.970967
1/3	12	73	145	0.057045	0.976275
1/2	18	108	110	0.054921	0.959158
2/3	25	145	73	0.04626	0.982807
3/4	28	162	56	0.039027	0.987569
5/6	31	183	35	0.039469	0.985794
8/9	33	194	24	0.037385	0.986167
11/12	34	200	18	0.0374	0.984802

Table S4. Details of training and testing samples, performances of capacity estimation models of Bayesian Ridge Regression.

Training samples/total samples	Number of training samples in every type	Number of training samples	Number of testing samples	RMSE	R ²
1/12	3	17	201	0.140528	0.85647
1/9	4	23	195	0.130806	0.879804
1/6	6	35	183	0.094032	0.936462
1/4	9	53	165	0.077236	0.958831
1/3	12	71	147	0.074801	0.961456
1/2	18	108	110	0.073269	0.962432
2/3	25	147	71	0.071622	0.962977
3/4	28	165	53	0.071107	0.962863
5/6	31	183	35	0.069141	0.963603
8/9	33	195	23	0.068834	0.960672
11/12	34	201	17	0.06772	0.961295

Table S5. Details of training and testing samples, performances of capacity estimation models of Gaussian Process Regression.

Training samples/total samples	Number of training samples in every type	Number of training samples	Number of testing samples	RMSE	R ²
1/12	3	18	200	0.078961	0.950387
1/9	4	25	193	0.078642	0.95258
1/6	6	36	182	0.069595	0.962725
1/4	9	53	165	0.058829	0.974705
1/3	12	73	145	0.050401	0.981512
1/2	18	109	109	0.042526	0.986414
2/3	25	145	73	0.038367	0.988252
3/4	28	162	56	0.036176	0.98867
5/6	31	183	35	0.03386	0.99076
8/9	33	194	24	0.03403	0.987787
11/12	34	200	18	0.032119	0.990233

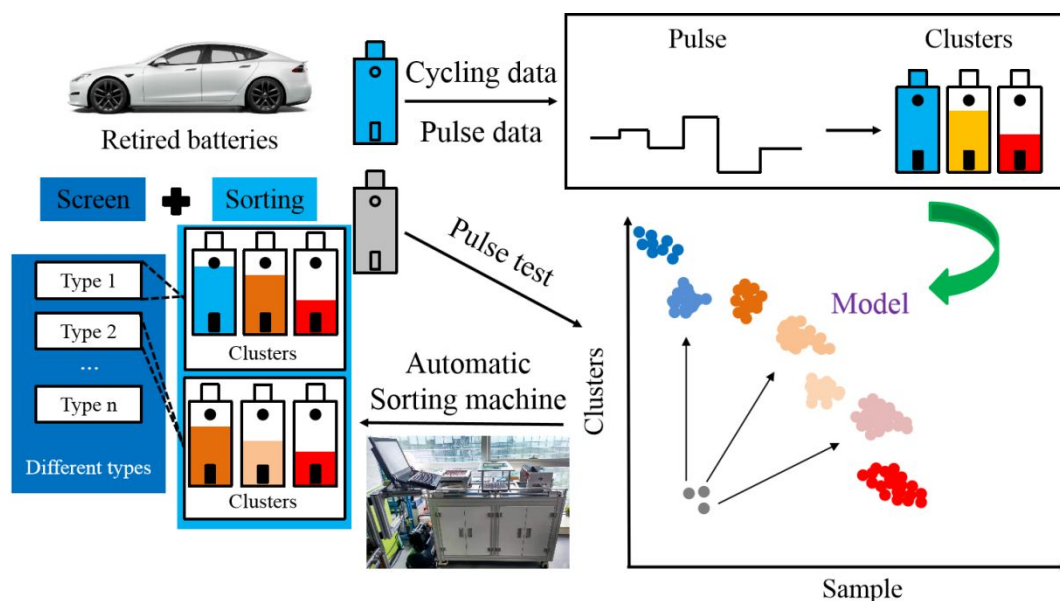


Fig. S12. Framework of sorting retired Lithium-ion batteries.

The framework of sorting retired lithium-ion batteries is shown in **Fig. S11**. In the secondary utilization of lithium-ion batteries, due to the performance of different batches of batteries from different manufacturers will be different, it is necessary to use Step 1 to screen LIBs into different types according to label, historical

information or other parameters. Step 2 is to establish the sorting model corresponding to different types of batteries. After collecting pulse data of retired LIBs, the automatic sorting machine can realize the rapid and accurate classification of LIBs with same performance. LIBs secondary utilization must be the same manufacturer, the same model, the same batch and the same performance.

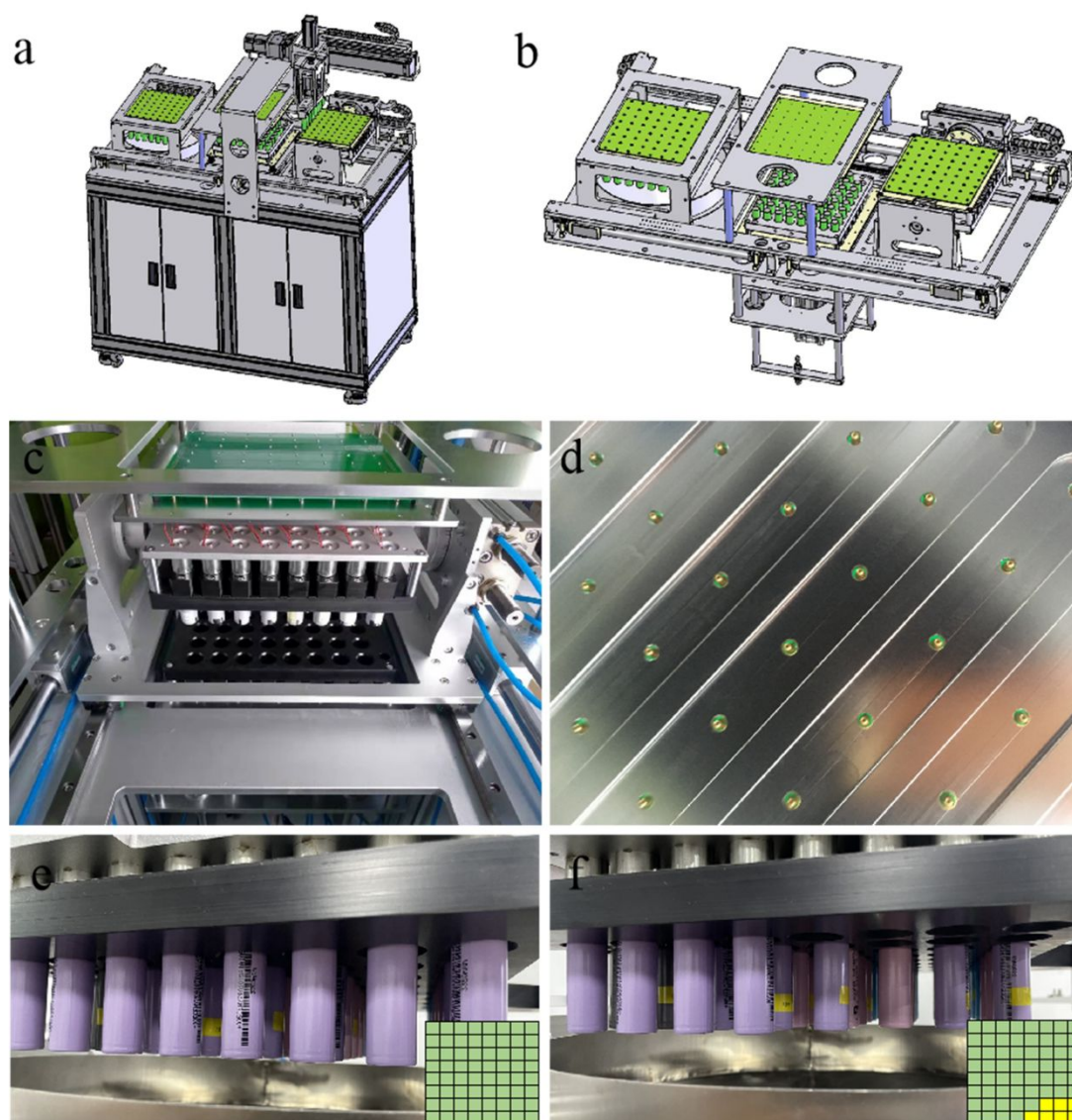


Fig. S13. Design and development of automatic battery sorting machine. (a) Design of automatic sorting machine, (b) Test core components of sorting machine; (c) Physical picture of sorting equipment. (d) Battery test probe array. (e) and (f) show the clustering module with the output of clustering results inserted. (e) shows the pristine state, seven batteries are clustered in (f).